GEEVI EVIDENCE BRIEF - PUBLISHED SOURCE - ATTRIBUTED SUMMARY

Beat AML 2018 - research-data and assay-provenance brief

Source type: Functional-genomic research cohort and dataset

Publication date: 2018-10-17

Citation: Tyner JW, Tognon CE, Bottomly D, et al. Functional genomic landscape of acute myeloid leukaemia. Nature. 2018;562:526-531. doi:10.1038/s41586-018-0623-z. https://www.nature.com/articles/s41586-018-0623-z

Question: How can AML clinical annotations, genomic findings, gene expression and ex vivo drug-response assays be linked without treating research assays as clinical recommendations?

Population: The 2018 publication describes 672 tumor specimens collected from 562 patients with AML. Multiple specimens can belong to one patient; specimen and patient denominators differ.

Intervention: Research profiling with whole-exome sequencing, RNA sequencing and ex vivo drug-sensitivity analyses.

Comparator: Research comparisons across molecular states and assay-response patterns; no randomized clinical treatment comparator.

Outcomes: Molecular and expression associations with ex vivo drug sensitivity Linked clinical, genomic, transcriptomic and functional data

Finding 1: The source reports associations between molecular features and laboratory drug-response measurements and makes a linked AML research dataset available through named repositories and Vizome. This is useful for designing research data relationships, including patient, specimen, assay, molecular result and analysis provenance. It does not show that an ex vivo result predicts clinical benefit for an individual patient.

Limitations: Publisher abstract, data-availability section, copyright metadata and the named dbGaP entry were verified. Subscription full text was not read. Observational and laboratory associations are not randomized evidence of treatment efficacy; missing assays, multiple specimens and assay context must remain explicit. The 2018 publication's dbGaP accession phs001657.v1.p1 is superseded. The old record lists 583 consented subjects, distinct from the publication cohort; preserve release and analysis-population provenance. No participant-level data were downloaded. Data-access permission and applicable dataset terms must be checked before ingestion or redistribution.

Applicability: Use the study to shape a separate research dataset model: patient-to-specimen links, collection time, assay method, quality status, clinical annotation, access permission and analysis version. Keep ex vivo sensitivity measurements separate from clinical observations and patient care decisions. Begin with a data dictionary and public aggregate examples. Do not import research participants as Geevi patients or label the demo a clinically validated matching tool.

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Original Geevi evidence brief, not the original article or a treatment recommendation.

Sources verified: 2026-10-11. Full-text appraisal and a current literature review remain necessary.